AI Prompt Engineering & Project Documentation

Candidate	SOHAIL A DODDAMANI
Position	AI Growth & Automation Associate (Work From Home)
Project	DoseDay: Medicine Reminder & Adherence Tracker
Area / domain	Website Development / Healthcare
Live app	https://doseday-project.netlify.app/
GitHub repository	https://github.com/sohailD18/doseday-medicine-reminder.git

1. Summary

DoseDay is a medicine reminder web app. Users add medicines with one or more daily times, tick doses off on a daily checklist, and review their adherence over 7 days, any month or a full year. Their history can be exported as a CSV file, which a Python script analyses to print compliance reports.

I used AI tools for planning and for generating the code, and I directed the requirements, tested the result and chose what to keep. The sections below list every prompt in order, with what the AI produced.

2. AI tools used and why

- **Gemini:** planning the project idea and a starting prompt roadmap.
- **Claude:** building and revising the code (index.html and adherence_report.py), explaining problems in simple words and giving exact find-and-replace edits.
- **Why these two:** Gemini gave me a quick project idea and a starting prompt roadmap. Claude let me describe problems in plain words, explained what was wrong, and gave exact find-and-replace edits and complete files, which is how I worked through each fix.

3. Prompt log

Prompts are quoted exactly as I typed them, including spelling. Text in brackets is my summary of a longer pasted message.

Step 1: Understanding the task and choosing an area

Tool	Claude
Goal	Understand the assessment PDF and choose a realistic project.
Exact prompt	> *I want to do this Website Development project so guide me what I need to do in simple words* > *(assessment PDF attached)*
What the AI produced	A step-by-step plan: pick a simple healthcare idea, choose AI tools, build in small steps, save every prompt, test, host on GitHub Pages, record the video and submit. Suggested a medicine reminder app or a clinic booking site.

Step 2: Gemini's project recommendation and base prompt

Tool	Gemini (shared with Claude)
Goal	Choose the project and get a starting prompt.

Exact prompt	*Act as an expert frontend web developer. I need to build a single-page Medicine Reminder Web Application for a job assessment. The app should be built using clean, modern HTML, Tailwind CSS (via CDN) for styling, and vanilla JavaScript.* *Features needed: 1. A clean, beautiful healthcare-themed dashboard header. 2. A form to add a medicine (Inputs: Medicine Name, Dosage, and Time/Frequency). 3. A beautiful 'Today's Schedule' checklist section where added medicines appear as cards or list items with a checkbox. 4. When a user checks a medicine, the text should visually strike through to show it's 'Taken'.* *Please provide the complete code in a single index.html file so it is easy to run.*
What the AI produced	Gemini recommended the Medicine Reminder app and supplied this prompt as Prompt 1. Claude reviewed it, noted that a basic checklist would look like many other submissions, and suggested extra features (saving data, edit and delete, progress bar, reminders, daily reset, medical disclaimer).

Step 3: First full build (Prompts 1 to 5 of the Gemini roadmap)

Tool	Claude
Goal	Get a working first version with all roadmap features.
Exact prompt	*Let's execute this exact 5-step prompt flow. Here is your roadmap to build the app and generate your Prompt Documentation at the same time.* *(followed by Gemini's five prompts, abridged: 1. base app; 2. localStorage persistence and disclaimer; 3. edit, delete and form validation; 4. progress bar and responsive layout; 5. code review and accessibility)*
What the AI produced	One index.html containing all five prompts' features: add form, daily checklist with strike-through, localStorage saving with a daily reset, edit and delete, validation messages, a progress bar, a disclaimer, ARIA labels and a responsive layout. Note: Claude built this consolidated version in one go rather than five separate runs.

Step 4: Deciding how to stand out

Tool	Claude
Goal	Decide whether to add login and register pages and how to make the project stronger.
Exact prompt	*Now my question is only page is enough or do you want to add login and register page also?* *I want to make this one best and stand out in all possible ways not has simple like this*
What the AI produced	Advised against login pages: in a frontend-only app they would be fake security. Recommended a 7-day adherence view, CSV export with a Python analysis script, multiple times per day, dark mode, demo data and a README.

Step 5: Upgraded build with the Python script

Tool	Claude
Goal	Add the stand-out features and the Python component.
Exact prompt	*Yes, absolutely build this upgraded version right now.* *(followed by Gemini's blueprint, abridged: 7-day adherence chart; CSV export button; Load Demo Data button; dark mode and multiple daily times; a separate Python script using the csv module to compute weekly compliance and print a terminal summary)*
What the AI produced	index.html (adherence chart, CSV export, demo data, dark mode, multiple times per day), adherence_report.py, README.md and a sample CSV. Claude's first attempt to save index.html failed because the file already existed; it moved the old version aside and wrote the new one.

Step 6: Console error

Tool	Claude
Goal	Understand and fix a browser console error.
Exact prompt	*Failed to load resource: the server responded with a status of 404 (Not Found)* *Now guide me with this dont change completely just guide me with find and replace*
What the AI produced	Explained it was the browser requesting a missing favicon and gave a one-line find-and-replace that adds an inline icon.

Step 7: Desktop layout and design

Tool	Claude
Goal	Make the layout correct on desktop and responsive on all devices, and improve the design.
Exact prompt	*For Desktop view the layout is not correct so guide me with responsive to all the devices and also make enhancement in the design*
What the AI produced	Twelve find-and-replace edits: wider page, two-column desktop layout (form left, schedule right, chart across the bottom), one column on tablets and phones, gradient header, softer cards. Claude applied them to the file first and checked that each text matched exactly once.

Step 8: Form section position

Tool	Claude
Goal	Stop the form section from staying fixed on screen.
Exact prompt	*form-title this section is absolute/ sticky make it relative to the page*
What the AI produced	Removed the sticky classes so the form scrolls with the page, with the single find-and-replace to do it.

Step 9: CSV date not visible in Excel

Tool	Claude
Goal	Understand why dates were not visible after exporting.
Exact prompt	*adherence_log.csv once i try to export i am unable to see the date in excel sheet now understand the issue then guide me other than dont change or effect anything*
What the AI produced	Explained that the dates are stored correctly but Excel's column is too narrow (it shows ######), with steps to widen the column and to check the file in Notepad. No code was changed.

Step 10: Notifications button and chart bar details

Tool	Claude
Goal	Fix the notify button and show details for each chart bar.
Exact prompt	*notify-btn this button is not working and guide me how to do* *inside this mt-4 flex items-end justify-between gap-2 h-44 sm:h-56 i have fill w-full rounded-t-md bg-teal-600 based on each i want to see the respective details here sched-title this section now give me full and correct one*
What the AI produced	Chart bars became clickable buttons showing each day's doses, and the notifications button gave visible messages about permission, secure pages and blocked states. Claude returned the whole corrected index.html.

Step 11: Schedule list follows the chart

Tool	Claude
Goal	Make the schedule list show the day selected in the chart.
Exact prompt	*it is not clickable notify-btn* *see here day-detail how data is shown with respect to chart day i selected* *i want the same respective details should be automatically happen here list understand the issue then guide me*
What the AI produced	The Today's schedule list now shows the selected day (read-only for past days, with a Back to today button), and the notify button was restyled as a visible button with a message below it. Claude tested this in a headless browser.

Step 12: Removing the duplicate panel

Tool	Claude
Goal	Remove duplicated information and keep the notification fix.
Exact prompt	*notify-btn this button is not working* *and here day-detail and sched-title there are same data shown twice make it at onces which one is best remove another one other than dont change or effect anything*
What the AI produced	Removed the day-detail panel and kept the schedule list. The notifications button now confirms when it is on and explains how to check device settings if no popup appears.

Step 13: Month and year views

Tool	Claude
Goal	Add month and year views to the adherence chart.
Exact prompt	*Now it is showing only 7-day adherence but what if i want for month and year like that* *(follow-up) For month i want perticuler month need to be selected and for year i need to see all the months once i click on any month respective to that month i want to see the dates data then like our existing logic*
What the AI produced	7 days / Month / Year tabs: a month picker with previous and next arrows, a year view of 12 months, click a month to open its days, click a day to see its schedule. Demo data now covers a year.

Step 14: Final files

Tool	Claude
Goal	Update the supporting files and prepare this documentation.
Exact prompt	*okay now give me other updated details of adherence_log README and adherence_report files and also as per requirement Prompt documentation: attached (PDF)*
What the AI produced	Updated README, a Python script with --days and --month options, a longer sample CSV and this document.

Step 15: Scrollable list, per-dose delete and no duplicates

Tool	Claude
Goal	Handle long lists, delete only one dose, and stop duplicate medicines.

Exact prompt	*sched-title make this section scrollable if i add many data or present there* *and while i delete i need to delete only respective data inside it sched-title section* *and also dont allow to add duplicated medician here sched-title whenever i try to create new one here form-title*
What the AI produced	The schedule list now scrolls after about six doses. The trash icon removes only the dose that was clicked (the medicine is removed only when it was its last time), with a confirmation that names the dose. Adding a medicine whose name already exists is blocked with a message under the name field. Claude tested adding, duplicate blocking and deleting in a headless browser.

Step 16: Updating only the changed files

Tool	Claude
Goal	Get updated files only where something changed.
Exact prompt	*okay now give me other updated details of adherence_log README and adherence_report files* *and also as per requirement Prompt documentation: attached (PDF)* *only changeable file not unchangd files*
What the AI produced	Updated README.md and this documentation. adherence_report.py and the sample CSV had no changes, so they were left as they were.

4. Other guidance I asked for (no code)

- Whether the project fits the job description and the "Full Stack Developer" tag; the answer was that the listing asks for web, Python, scraping and AI skills, and that this project is a frontend app plus a Python script, not full stack.
- What each file in the repository is for, and how to prepare the video, email and covering letter.
- A review of a draft of this documentation, which led me to remove claims I had not actually tested.

5. My own testing

I tested the app and script myself. Results:

Test	What I did and what I saw
Load demo data, then export adherence_log.csv	I exported the CSV and opened it in Excel, but could not see the dates at first. Claude explained that the date column was too narrow for Excel to display the dates, and gave steps to widen it.
Run the Python script: python adherence_report.py adherence_log.csv (also with --days 30 and --month)	Got Output.
7 days, Month and Year views; click a month, then a day; check the schedule list changes	With the 7-day chart I noticed the details panel and the schedule list showed the same information twice, so I asked for one to be removed. I also asked for month and year views once I saw only 7 days.
Submit the form with empty fields	A New Card was added.
Dark mode, then refresh the page	It stays in Dark Mode after Refreshing the page.
Phone layout (device mode or a real phone)	Its Working Fine.
Notifications button on the live GitHub Pages link	The notifications button seemed not to respond when I clicked it, and I reported it several times. Claude changed it to a visible button with a message below it that explains the permission state.

Test	What I did and what I saw
Reminder alert at a set time with the page open	Its Working Fine.
Add 10 or more medicines, try adding the same name twice, and delete one dose of a medicine with two times	I noticed that deleting removed all of a medicine's times and that I could add the same medicine twice, so I asked for these to be fixed. Its Working Fine.

6. My decisions, problems and next steps

Why I kept the app frontend-only: I kept it to HTML, CSS and JavaScript plus a Python script so it is quick to host and easy for reviewers to open. I decided against login and register pages because, without a server, a login would only be for show and would not protect anything. Keeping data in the browser also keeps health information on the user's own device.

Problems I found and how I handled them: (1) The browser console showed a 404 error. I asked Claude, which explained it was a missing tab icon and gave a one-line fix. (2) The desktop layout looked wrong, a single narrow column on a wide screen. I asked for a responsive layout and applied the find-and-replace edits for a two-column desktop view. (3) The form section stayed fixed on screen, so I asked for it to scroll with the page. (4) The notifications button did not seem to work. It took three rounds, and the app now shows a message explaining the permission or secure-page problem. (5) The schedule list and the details panel showed the same data, so I had one removed and made the schedule list follow the chart. (6) Dates in the exported CSV looked missing in Excel because the column was too narrow. (7) The list became long with many medicines, deleting removed all of a medicine's times, and the same medicine could be added twice, so I asked for a scrollable list, per-dose delete and duplicate blocking.

What I would improve next: Add a small backend so reminders can arrive in the background and data can sync across devices, send weekly adherence summaries by email or WhatsApp using the CSV and Python script, and add a way to pick any month in the Python report from the app itself.

7. Checks run by the AI, and limitations

Claude reported running these checks while building, and I have not independently repeated all of them: a JavaScript syntax check; running the Python script on sample data and with invalid input; and headless-browser tests of the chart views, schedule list and notification logic.

- Reminders only fire while the page is open; there is no background service.
- Data is stored in the browser only (localStorage), about 400 days, with no accounts or server.
- Browser notifications need https or localhost and the user's permission.
- This app is a reminder tool and does not constitute medical advice.